

II-5 Delayed Reactivity Feedback Phenomena¹

Moderator Temperature Effect

Although feedback effects associated with the fuel are the quickest acting, those associated with the temperature of the moderator may be larger in magnitude. In thermal reactors, temperature changes in the moderator introduce changes in neutron multiplication through a variety of mechanisms. In a pressurized water reactor (PWR), expansion of the water leads to the expulsion of water from the core and thus to the loss of moderator. Since a PWR will generally be under-moderated, the loss of moderator will lead to a fall in k . Typically for a PWR the moderator temperature coefficient of reactivity is about -10^{-4} per $^{\circ}\text{C}$. On the other hand, thermal expansion of the graphite in a graphite-moderated reactor has an insignificant effect on the moderator temperature coefficient of reactivity. With graphite-moderated reactors the most important result of changing moderator temperature is that the kinetic energy of the atoms with which the neutrons achieve some form of equilibrium is changed. Thus the energy spectrum of neutrons in the reactor is changed. The variations of neutron cross-section with neutron energy are not the same for all core materials and thus changes in temperature will result in changes in the number of fission reactions taking place relative to other neutron reactions, such as capture and leakage. For graphite-moderated, uranium-fueled lattices the effect is fairly small and negative. With significant amounts of plutonium present, however, the situation is markedly different. The cross-section of the fissile isotope ^{239}Pu has a large resonance at 0.3 eV which is between 5 and 10 times the mean neutron energy. An increase in moderator temperature leads to an increase in mean neutron energy and thus to an increase in the number of absorptions (and hence fissions) in ^{239}Pu . Consequently the moderator temperature coefficient becomes positive and increases in magnitude with the addition of plutonium to the fuel. This trend does not increase indefinitely, since with high concentration of plutonium there is already a large fraction of the neutrons absorbed in ^{239}Pu and increases in resonance absorption must be at the expense of more efficient absorption (higher fission/absorption ratio) at lower energies.

With natural uranium fueled (for example, Magnox reactors) the moderator temperature coefficient of reactivity with unirradiated fuel is roughly $\alpha_m = -2 \times 10^{-5}$ per $^{\circ}\text{C}$. Usually, in these reactors fuel is discharged at about 3000 MWD/te with on-load refueling schemes. A moderator temperature coefficients of reactivity in the region of $\alpha_m = +10 \times 10^{-5}$ per $^{\circ}\text{C}$ arise after 2 or 3 years operation. At first sight, since this positive moderator temperature coefficient (and resulting positive feedback) is much greater than the magnitude of the negative fuel temperature coefficient, it might be thought that serious difficulties in control would result. It is explained later in this chapter that the situation is alleviated because fuel temperature changes following power changes are both considerably larger and much more rapid than the moderator temperature changes. Nevertheless the existence of a significant positive feedback effect is of great importance in the graphite-moderated reactor and as we shall see has in many instances determined the type of control system required for this reactor. Because quite accurate estimates of moderator temperature coefficients are required as reactors are designed, much theoretical and experimental work has been devoted to formulating methods of arriving at such estimates by calculation. Relevant calculations require accurate techniques for computing the neutron energy spectrum, covering its dependence on position in the lattice cell, and these in turn rely for accuracy on detailed knowledge of nuclear data involving both the neutron-absorbing

¹ Source:

1. Ott & Neuhold, *Introductory Nuclear Reactor Dynamics*, American Nuclear Society, 1985

characteristics of the fuel and the neutron-scattering properties of the moderator. The net effect is the result of several competing processes and in general fairly sophisticated calculations are necessary to obtain an accuracy of say 1×10^{-5} per °C. These calculations are normally compared with measurements in clean, heated lattices before application to the power reactor situation. Experimental values for the coefficient may also be derived from experience on specially designed experiments in power reactors, although in this case the accuracy may not be so high.

In a manner analogous to the fuel temperature coefficient we thus have a moderator coefficient of reactivity given by

$$\Delta k = \alpha_m(T_m) \Delta T_m \quad (\text{II-154})$$

Changes in the moderator temperature ΔT_m are related to changes in the neutron flux $\Delta \phi$ by a suitable set of heat-transfer equations for the fuel lattice. The simplest type is of the form discussed above for the fuel:

$$\tau_m \frac{\partial}{\partial t} \Delta T_m = \frac{\Delta \phi}{\phi^*} T_m^* - \Delta T_m \quad (\text{II.155})$$

For graphite lattices the moderator time constant, τ_m , is much larger than the fuel time constant, τ_f , due to the large thermal capacity of the moderator bricks. Please note that T_m^* and ϕ^* are the initial values of moderator temperature and flux. The coupling between moderator temperature and neutron flux in Magnox reactors is fairly strong. In pressure tube reactors (CANDU), however, the moderator is thermally insulated from the reactor coolant and is maintained near constant temperature (under 100°C) by itself being circulated through a heat exchanger. Consequently the coupling between neutron flux and main moderator temperature is weak. Coupling is also weak in the gas-cooled reactors, where sleeves around the fuel element thermally insulate the moderator from the coolant flowing over the fuel. Weak coupling would appear in Eq. (II.155) as a low value for T_m^* .

Voidage Effects

These are important in the pressure vessel boiling water reactor (BWR) and in the pressure tube steam-generating heavy-water reactor (SGHWR). The steam is passing to the turbine being generated in the reactor core in both these types. The effects of steam voidage changes on multiplication in these reactors are somewhat similar to the effects of coolant temperature changes in a PWR in that both cause changes in moderator density. These changes in the number of water molecules affect slowing down fission, capture, and leakage properties of the reactor lattice and the net effect is the result of a complex interaction between them all, which may have either sign.

The response time of voidage to changes in power is a matter of only a few seconds and it is desirable to have a small negative void coefficient of reactivity so that the control system does not have to stabilize reactor power on a short time scale. On the other hand, too large a negative void coefficient could cause oscillatory instabilities and might lead to the need for rather high reactivity rates from the reactivity control system in order to achieve the highest rates of change of electrical output demanded in normal operation of the plant.

A useful negative void coefficient may be achieved by under-moderating the reactor. This reduction in the ratio of moderator to fuel volumes tends to reduce the size and thus the capital

cost of the reactor but this at the expense of an inferior neutron economy leading to a higher required fuel enrichment. On balance the net economical effect is very small.

The Effect of Temperature on Power Transients

As our first illustration we shall consider how the temperature feedback mechanisms affect a reactor power transient. We shall assume that the reactor flux shape remains unchanged throughout the transient so that $\nabla^2 \phi$ is proportional to ϕ for all values of t and special effects are completely ignored. We neglect also all delayed neutron effects. Suppose that a reactor is running steadily at power when the control rods are moved slightly to cause a small increase in multiplication δk . For a few seconds the effects of the change in power on reactivity will be insignificant. Subsequently, however, the rising fuel temperature will begin to reduce the net reactivity because of the negative fuel temperature coefficient of reactivity. The fall in net reactivity will arrest the rise in power and after about a few seconds to a minute (determined by the time constant of the fuel τ_f) the power will stabilize, at least temporarily, at such a level that the drop in reactivity due to increase in fuel temperature compensates for the initial reactivity excursion. At this stage the equation

$$\delta k + \alpha_f \Delta T_f = 0 \quad (\text{II.156})$$

is obeyed to a fair approximation, ΔT_f being the change in statistically weighted mean fuel temperature and α_f the fuel temperature coefficient of multiplication. Now if we neglect the time constant of fuel and coolant and neglect changes in the temperature of the coolant gas entering the core (if considering gas-cooled reactors), the heat-transfer equation for the fuel reduces simply to

$$\Delta T_f = T_f^* \frac{\phi'}{\phi^*}, \quad (\text{II.157})$$

where $\alpha_f T_f^*$ is the (constant) mean value of the fuel temperature measured above coolant inlet and ϕ' is the change in neutron flux. From the Eqs. (II.156) and (II.157) we have the relative power change

$$\delta P = \frac{\phi'}{\phi^*} = -\frac{\delta k}{\alpha_f T_f^*},$$

which is positive for positive δk since in practice $\alpha_f T_f^*$ will always be negative.

The increase in power will now cause moderator temperatures to begin to rise and succeeding events will depend on the sign and magnitude of the moderator temperature coefficient of reactivity. The time lag between power changes and changes in mean moderator temperature in LWRs is between 3-4 seconds, while in the Magnox type of reactor is between 5 and 10 min and we denote it here by the symbol τ_m . We rewrite Eq. (II.155)

$$\tau_m \frac{d}{dt} \Delta T_m + \Delta T_m = T_m^* \delta P, \quad (\text{II.158})$$

where T_m^* is the mean graphite temperature measured above gas inlet temperature. Eq. (II.157) still holds but the reactivity balance equation now generalizes to

$$\delta k + \alpha_f \Delta T_f + \alpha_m \Delta T_m = 0, \quad (\text{II.159})$$

where α_m is the mean moderator temperature coefficient of reactivity. Eqs. (II.157), (II.158), and, (II.159) give now

$$\alpha_f \Delta T_f^* \tau_m \frac{d}{dt} \delta P + (\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*) \delta P = -\delta k,$$

The solution of this equation is

$$\delta P = -\frac{\delta k}{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)} \left\{ 1 + \frac{\alpha_m \Delta T_m^*}{\alpha_f \Delta T_f^*} e^{-\omega t} \right\}, \quad (\text{II.160})$$

where

$$\omega = \frac{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)}{\alpha_f \Delta T_f^* \tau_m}.$$

For small values of t , δP is close to the value $-\frac{\delta k}{\alpha_f \Delta T_f^*}$. As time increases, however, the behavior of δP depends critically on the value of ω . For positive values of ω the magnitude of δP moves away from $-\frac{\delta k}{\alpha_f \Delta T_f^*}$ towards the value $-\frac{\delta k}{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)}$

which will entail an increase or decrease in δP depending upon the sign of α_m . For negative values of ω , however, the power behavior is quite different. The exponential term $e^{-\omega t}$ increases indefinitely as t increases so that δP increases without limit. In these circumstances we say that the power output of the reactor is unstable. The condition for instability is clearly $\omega < 0$.

Since $\alpha_f \Delta T_f^*$ is always negative in practice, the instability criterion may also be written

$$\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^* > 0. \quad (\text{II.161})$$

This is an important result and illustrates the fact that it is the weighted sum of the feedback (reactivity) coefficients, which determines whether or not the net feedback is positive. The weighting depends upon the strength of the coupling between temperature and power which is reflected in the steady state material temperatures. The fuel temperature is more sensitive function of neutron flux or power since it changes more in a transient. It thus has a higher weighting.

In Fig. II.14 we show the variation of δP with time (using Eq. II.159) for the case of a reactivity step $\delta k = 10^{-4}$ for a reactor with $\tau_m = 10$ min, $\alpha_m = -2 \times 10^{-5}$ per °C, $T_f^* = 250$ °C and $T_m^* = 130$ °C. Since we assume $\tau_f = 0$, there is an initial instantaneous power rise of 2 per cent. The subsequent motion is stable if $\alpha_m < 3.85 \times 10^{-5}$ per °C.

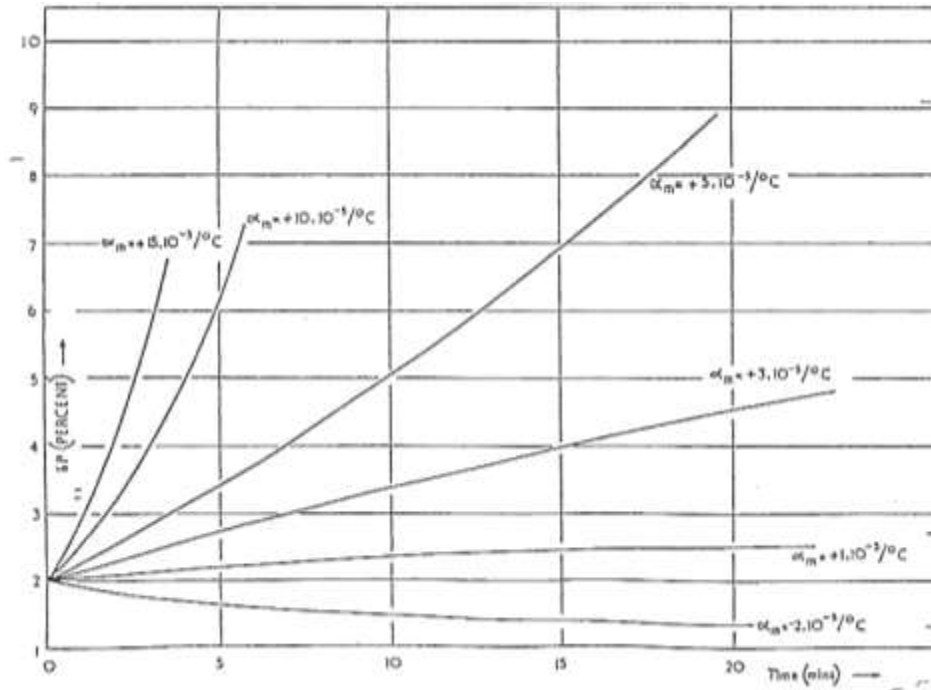


Fig. II-14. Percentage power change for case $\delta k = 10^{-4}$; $\tau_f = 0$; $\tau_m = 10$ min; $\alpha_m = -2 \times 10^{-5}$ per $^{\circ}\text{C}$; $T_f^* = 250^{\circ}\text{C}$ and $T_m^* = 130^{\circ}\text{C}$.

General Reactor Stability

Combining all the feedback effects discussed above, we are now in a position to study the relevant set of power reactor equations

$$l \left[\frac{\partial \phi}{\partial t} + \sum_i \frac{\partial \phi_i}{\partial t} \right] = M^2 \nabla^2 \phi + (k - 1) \phi + \delta k \phi$$

$$l \frac{\partial \phi_i}{\partial t} = k \beta_i \phi - \lambda_i l \phi_i$$

$$k = k^* + \alpha_f \Delta T_f + \alpha_m \Delta T_m + \alpha_x \Delta x + \dots$$

together with suitable auxiliary equations describing T_f , T_m , x , ... in terms of ϕ . The solution of this set of coupled, non-linear differential equations for given initial conditions and disturbances provides, of course, a very formidable problem.

We turn now to a general discussion of the stability and other broad characteristics associated with the solution of the power reactor equations. This set of coupled, non-linear, differential

equations presents a very difficult problem - so difficult, in fact, that an analytical (or near-analytical) treatment seems hardly feasible. Some attempt has, however, been made to study the stability of rather simpler sets of non-linear equations. No attempt is made here to reproduce these arguments. The reader interested in the general characteristics of non-linear differential equations must refer to the relevant literature.

What we can consider here, however, is the stability of the set of equations in the small perturbation. That is, we can consider small perturbations to an equilibrium steady-state solution and investigate in what circumstances the perturbations will grow or subside. Consideration of only small disturbances enables us effectively to linearize the relevant equations since we may neglect products of small quantities.

We suppose, therefore, that the various time-dependent quantities receive small perturbations from their steady-state values, i.e.

$$\phi = \phi^* + \phi'$$

Then neglecting the product of small (primed) quantities we have

$$l \left[\frac{\partial \phi'}{\partial t} + \sum_i \frac{\partial \phi_i'}{\partial t} \right] = M^2 \nabla^2 \phi' + (\alpha_f T_f' + \alpha_m T_m' + \dots) \phi^* + (k^* - 1) \phi' + \delta k (\phi^* + \phi'),$$

$$l \frac{\partial \phi_i'}{\partial t} = (\alpha_f T_f' + \alpha_m T_m' + \dots) \beta_i \phi^* + k^* \beta_i \phi_i' - \lambda_i l \phi_i'$$

together with heat-transfer equations for fuel, moderator and coolant

$$\begin{aligned} \frac{\partial T_f'}{\partial t} &= K_f \phi' - (T_f' - T_c') \gamma_f \\ \frac{\partial T_m'}{\partial t} &= K_m \phi' - (T_m' - T_c') \gamma_m \\ V \frac{\partial T_c'}{\partial z} &= (T_f - T_c) \gamma_{fc} + (T_m' - T_c') \gamma_{mc} \end{aligned} \quad (\text{II.162})$$

In order to investigate the time behavior of this set of equations we express the flux perturbations as a sum of terms, each separable in time and space. That is, we set

$$\phi' = \sum_m A_m a_m(t) h_m(r, \theta, Z) \quad (\text{II.163})$$

Ideally we should choose the h_m functions to be the normal modes of the set of equations (II.162) in the same way as the simple neutronics equations without feedback are treated. The time dependence associated with these modes is of the form

$$a_m(t) = e^{\omega_m t}$$

The values of ω_m are determined only by the equations themselves and not by the nature of the initial disturbance. The initial conditions, however, determine the values of A_m - i.e. the magnitudes of the modal components.

Our objective in what follows will be to discover in what circumstances any particular value of ω_m is positive (or has positive real part). For these circumstances that particular modal component will increase following a disturbance and since a general disturbance will contain contributions from all modes it may be concluded that the system is unstable.

In practice it is extremely difficult to obtain the normal modes of this set of equations. A purely numerical approach would be necessary and this would be inconsistent with our desire to obtain some physical feeling for the results. Instead we choose a set of harmonics, which we think will be very similar to the normal modes, but are relatively simple to handle analytically. The harmonics we shall choose are in fact given by the eigenvalue equation

$$\nabla^2 F_m + \frac{(k^* - 1)}{M^2} F_m + \mu_m F_m = 0 \quad (\text{II.164})$$

The fundamental ($m=0$) harmonic is associated with the eigenvalue $\mu_0 = 0$ and is clearly proportional to the steady-state flux level ϕ^* . The higher harmonics are related to the normal modes of the neutronics equation with no feedback. Thus when the feedback is reasonably small we may anticipate that the chosen harmonics are fairly close approximations to the actual modes of the reactor system. The fact that they are not identical with them will be revealed by the presence of cross-coupling between the chosen harmonics. The magnitudes of these terms may be investigated and are normally found to be small. In what follows we shall ignore all such cross-coupling terms.

For a cylindrical reactor with a constant value $\frac{(k^* - 1)}{M^2}$ the harmonics are given simply by

$$F_m \equiv F_{l\gamma n} = A_m J_\gamma(\beta_{\gamma l} r) \cos \gamma \theta \sin \frac{n\pi z}{L} \quad (\text{II.165})$$

where J_γ denotes a Bessel function of order γ and the quantities l, γ, n take integer values only. The suffix l indicates the particular zero of the Bessel function considered and L is the reactor (extrapolated) height. The eigenvalues of Eqs. (II.164) are given by

$$\frac{n^2 \pi^2}{L^2} + \beta_{\gamma l}^2 = \frac{k^* - 1}{M^2} + \mu_m. \quad (\text{II.166})$$

The fundamental harmonic ($m = 0$) is given by

$$l = 1; \quad \gamma = 0; \quad n = 1.$$

The first azimuthal harmonic ($l = 1; \gamma = 1; n = 1$) will be denoted by $m = 1$, the second azimuthal harmonic ($l = 1; \gamma = 2; n = 1$) by $m = 2$, and so on. In general the radial azimuthal harmonics of interest will be written in the form

$$F_m \equiv f_m(r, \theta) g_0(z),$$

where

$$g_0(z) = \sin\left(\frac{\pi z}{L}\right)$$

is the steady-state axial flux form. In what follows we shall restrict attention to a consideration of the f_m component. Using the condition that f_m should vanish at the reactor boundary we have from Eq. (II.166) and the properties of Bessel functions

$$\begin{aligned}\mu_0 R_0^2 &= 0, \\ \mu_1 R_0^2 &= 8.90, \\ \mu_2 R_0^2 &= 20.59, \text{ etc.},\end{aligned}$$

where R_0 is the (extrapolated) reactor radius. Returning now to the expansion of the flux perturbation, we may write

$$\phi' = \sum_m A_m e^{\omega_m t} f_m(r, \theta) g_0(z). \quad (\text{II.167})$$

Further, since the delayed neutrons, ^{135}Xe and ^{138}I distributions are local functions of the neutron flux we may write also

$$\begin{aligned}\phi_l' &= \sum_m B_m e^{\omega_m t} f_m(r, \theta) g_0(z) \\ i' &= \sum_m C_m e^{\omega_m t} f_m(r, \theta) g_0(z) \\ x' &= \sum_m D_m e^{\omega_m t} f_m(r, \theta) g_0(z)\end{aligned} \quad (\text{II.168})$$

We now substitute Eqs. (II.167) and (II.168) into Eq. (II.162), multiply by F_n , and integrate over the reactor core. Since we are concerned only with the ω_m we omit the applied multiplication δk . We also neglect cross-coupling between the harmonics and obtain

$$l\omega_n + k_n^* \omega_n \sum_i \frac{\beta_i}{\omega_n + \lambda_i} - \alpha_{Xk} \phi_n^* \Gamma_n + M^2 \mu_n - \int_V (\alpha_f T_n' + \alpha_m T_m') \phi^* f_n g_0 dV = 0 \quad (\text{II.169})$$

where

$$Y_n^* = \int_V Y^* f_n^2 g_0^2 dV$$

and

$$\Gamma_n = \frac{\sigma_X (1-s) - x_n^* + \frac{\lambda_I s}{\omega_n + \lambda_I}}{\omega_n + \lambda_X + \sigma_X \phi_n^*} \quad (\text{II.170})$$

is the xenon transfer function and we have included only ^{135}Xe of the fission product poisons.

The evaluation of the integral occurring in Eq. (II.169) requires a consideration of the heat-transfer equation. Before doing that we make one further assumption - that the heat required to change the temperature of the various components is negligible in comparison with the heat removed from the reactor, i.e.

$$C_f \frac{\partial T_f}{\partial t} + C_m \frac{\partial T_m}{\partial t} + C_c \frac{\partial T_c}{\partial t} \ll VC_c \frac{\partial T_c}{\partial z}.$$

With this assumption we find that

$$\int_V T_f' \phi^* f_n g_0 dV = \frac{\gamma_f T_{fn}^*}{\omega_n + \gamma_f} \quad (\text{II.171})$$

$$\int_V T_m' \phi^* f_n g_0 dV = \frac{\gamma_m T_{mn}^*}{\omega_n + \gamma_m}$$

where $\gamma_f = 1/\tau_f$, etc. Thus using Eqs. (II.169) and (II.171) the equation for the time constant of the n^{th} mode becomes

$$H_n(\omega_n) = 0, \quad (\text{II.172})$$

where

$$H_n(\omega_n) \equiv l\omega + k_n^* \omega \sum_i \frac{\beta_i}{\omega + \lambda_i} - \alpha_x \phi_n^* \Gamma_n + M^2 \mu_n - \frac{\gamma_f T_{fn}^*}{\omega_n + \gamma_f} - \frac{\gamma_m T_{mn}^*}{\omega_n + \gamma_m}. \quad (\text{II.173})$$

The time behavior of the reactor system for given conditions is thus known when Eq. (II.172) is solved. In fact Eq. (II.172) as written is of degree $I + 5$ where I is the number of delayed neutron groups considered. There are thus $(I + 5)$ separate values of ω_n (and correspondingly $(I + 5)$ values of A_n , B_n , etc.). As mentioned above, of particular interest is the condition that any of these $(I + 5)$ values of ω_n is positive (or has positive real part). For the fundamental mode ($n = 0$) equation (II.172) is a generalization of the inhour equation to take account of feedback effects.

The general form of Eq. (II.172) is of a rather higher degree than is convenient to handle. We may, however, make still further simplifying assumptions in order to highlight certain features. We may, for example, ignore delayed neutrons, or perhaps better consider only a representative group given by

$$k_n^* \sum_{i=1}^I \frac{\beta_i}{\omega + \lambda_i} = \frac{\beta}{\omega + \lambda}.$$

We may also ignore the xenon effect in order to study the particular role of the heat-transfer feedback (this would be appropriate to a core state immediately after start-up) and vice versa.

Stability Condition

Returning to Eq. (II.172) we may express the equation for ω_n in the form

$$\sum_m B_{nm} \omega_n^m = 0. \quad (\text{II.174})$$

where B_{nm} are simply coefficients obtained from Eq. (II.173). Now when all the B_{nm} are positive there can be no real root of Eq. (II.174), which is greater than zero. The conditions for complex roots are given by the Routh-Hurwitz criteria, but it may be shown that in our particular case there are no complex roots with positive real part if $B_{nm} > 0$.

The condition

$$B_{nm} > 0, \quad \text{all } m \quad (\text{II.174})$$

is thus sufficient to establish the stability of the steady-state reactor to perturbations entering through the n^{th} harmonic. In fact the critical coefficient turns out to be B_{n1} , and the stability condition is

$$M^2 \mu_n - (\alpha_f T_{fn}^* + \alpha_m T_{mn}^*) + \alpha_x (x_n^* - 1 + s) \frac{\phi_n^* \sigma_x}{\lambda_I + \lambda_x + \phi_n^* \sigma_x} > 0 \quad (\text{II.175})$$

The inequality (II.175) illustrates the fact that the question of reactor stability is determined by the competition between neutron leakage, heat transfer effects, and ^{135}Xe fission product poison.

The neutron leakage term is never destabilizing. In the case of the fundamental harmonic ($n = 0$) it is zero, but for higher harmonics the term is stabilizing and increases in magnitude with increasing n . For the temperature effect it is seen that what is important is the product of the temperature coefficient with the mean steady-state temperature of the material measured above coolant inlet temperature. This temperature is a measure of the efficiency of heat removal from the material. Thus when T^* small, heat is removed efficiently and a change in neutron flux does not lead to appreciable material temperature change. This temperature effect alone is clearly stabilizing provided that

$$(\alpha_f T_{fn}^* + \alpha_m T_{mn}^*) < 0$$

This is just the condition that we met earlier in the chapter.

The xenon term is always destabilizing (since α_x is negative). Its magnitude depends upon the steady-state neutron flux level through the mean reactivity held by ^{135}Xe and the ratio of the forced to the total decay rate of xenon and iodine. It is relatively insensitive to the harmonic number n . At high flux level the magnitude saturates at the value $\alpha_x s$.

In a graphite-moderated power reactor only the coefficient α_m varies significantly with time; this is due to the build-up of plutonium isotopes in the fuel. It is thus convenient to express the stability criterion in terms of the moderator temperature coefficient of reactivity. The condition for stability is then

$$\alpha_m T_{mn}^* < \frac{M^2}{R_0^2} \mu_n R_0^2 + Q_n - \alpha_f T_{fn}^* \quad (\text{II.176})$$

where the xenon term is

$$Q_n = \frac{\alpha_x (x_n^* - 1 + s) \phi_n^* \sigma_x}{\lambda_I + \lambda_x + \phi_n^* \sigma_x}.$$

In a typical Magnox reactor $Q_n = -6 \times 10^{-3}$ and the fuel term is given by $\alpha_f T_{fn}^* = -5 \times 10^{-3}$. The two nearly cancel. Thus we see that for stability in the fundamental mode a small and even negative value for α_m is required.

Spatial Instabilities

It may be seen from the stability criterion that, for a given reactor, instability first enters through the fundamental harmonic and then, with increasing α_m , through higher and higher harmonics. This is basically due to the fact that for a non-fundamental harmonic perturbation neutrons are

able to leak or diffuse from the regions of increased neutron density to those regions where the density has fallen below normal. This stabilizing effect becomes more pronounced as the number of such regions of rising and falling neutron density - and hence the harmonic number - increases. The xenon and temperature feedback terms, on the other hand, are relatively insensitive to the harmonic number.

For a given harmonic, the onset of instability is enhanced by increasing power rating (i.e. neutron flux), reactor size, temperature coefficient of reactivity and, for positive coefficients, temperature rise through reactor core. This is illustrated in Table II-V, where the critical value of $\alpha_m T_{mn}^*$ is given for the first azimuthal harmonic ($n = 1$) for the case $\alpha_f T_{fn}^* = -5 \times 10^{-3}$ in a uniform, natural uranium reactor.

TABLE II-V

Critical Values of a $\alpha_m T_{m1}^* \times 10^3$ for Stability in the First Azimuthal Mode of a Uniform Reactor with $\alpha_f T_{fn}^* = -5 \times 10^{-3}$

M^2 / R^2	Maximum rating (MW/te)						
	2	4	6	8	10	20	50
1×10^{-3}	11.4	9.4	7.9	6.8	5.9	3.5	1.6
2×10^{-3}	19.3	17.3	15.8	14.7	13.8	11.4	9.5
5×10^{-3}	46	44	42.5	41.4	40.5	38.1	39.2

The variation of the stability threshold with harmonic is shown in Table II-VI for the particular case of $M^2 / R_0^2 = 2 \times 10^{-3}$, and Maximum rating = 4 MW/te.

TABLE II-VI

Critical Values of a $\alpha_m T_{mn}^*$ for Stability in the First Azimuthal Mode of a Uniform Reactor with $\alpha_f T_{fn}^* = -5 \times 10^{-3}$, $M^2 / R_0^2 = 2 \times 10^{-3}$, Maximum rating = 4 MW/te

Fundamental	-0.7×10^{-3}
First azimuthal	17.3×10^{-3}
Second azimuthal	40.7×10^{-3}
First radial	48.9×10^{-3}

Time Constant of Perturbed Motion

The equation (II.174), in principle, enables all the time constants of the perturbed motion to be determined. In general the time constant of greatest interest (i.e. with largest real part) is complex, i.e. $\omega_n = \omega_{n1} + i\omega_{n2}$. The time behavior of the neutron flux perturbation after more rapid transients (smaller values of ω_{n1} have decayed) is thus oscillatory and given by

$$\phi' \sim e^{\omega_{n1} t} \cos(\omega_{n2} t + \delta).$$

Solutions of small real part are given by an approximate form of Eq. (II.174) in which only the first three terms are retained, viz.

$$\sum_{m=0}^2 B_{nm} \omega_n^m = 0,$$

and we anticipate that this equation for w is valid in the neighborhood of the stability threshold. This quadratic equation yields the general solution (for $\omega = \omega_1 + i\omega_2$)

$$\frac{-2\omega_1}{\lambda_x} = S_n = \frac{L_n \left(1 + \frac{\lambda_I}{\lambda_x} + \frac{\phi_n^* \sigma_x}{\lambda_x} \right) + \alpha_x \frac{\phi_n^* \sigma_x}{\lambda_x} (x_n^* - 1 + s)}{L_n + \alpha_x \phi_n^* \sigma_x (x_n^* - 1 + s)} \quad (\text{II.177})$$

$$\frac{4\omega_2^2}{\lambda_x^2} = R_n = \frac{4\lambda_I}{\lambda_x} \left[S_n - \frac{L_n \lambda_I / \lambda_x + \alpha_x \phi_n^* \sigma_x / \lambda_x}{L_n + \alpha_x \phi_n^* (x_n^* - 1 + s) \tau_m} \right] - S_n^2, \quad (\text{II.178})$$

where

$$L_n = M^2 \mu_n - (\alpha_f T_{fn}^* + \alpha_m T_{mn}^*).$$

For a reactor on the threshold of instability ($\omega_1 = 0$) the motion is oscillatory and the period of oscillation is given by

$$R_n = \frac{4\lambda_I / \lambda_x}{x_n^* - 1 + s} \left[\frac{s(1 + \phi_n^* \sigma_x / \lambda_x) + \lambda_I (1 - x_n^*) / \lambda_x}{1 - \lambda_x \tau_m (1 + \lambda_I / \lambda_x + \phi_n^* \sigma_x / \lambda_x)} \right]. \quad (\text{II.179})$$

Typically the period lies in the range 24-30 h, and is clearly determined by the xenon-iodine time characteristics. The perturbed motion ceases to be oscillatory when $R_n \leq 0$. This will occur whenever S_n is sufficiently large and negative, i.e. when the reactor is sufficiently over the threshold of instability. In this case the behavior of the reactor is dominated by the non-oscillatory and fairly rapidly-acting temperature effects. The variations in time constants with increasing α_m are illustrated in Fig. II-15 for the first azimuthal harmonic.

The case considered is a typical Magnox reactor with $M^2 / R_0^2 = 1.7 \times 10^{-3}$, $\alpha_f T_{f1}^* = -5 \times 10^{-3}$ and Maximum rating = 4 MW/te.

It will be observed that for this harmonic, instability enters $\alpha_m T_{m1}^* = 8 \times 10^{-3}$. For $8 \times 10^{-3} < \alpha_m T_{m1}^* < 13 \times 10^{-3}$ the instability is oscillatory in nature ($R_n > 0$) with periods of at least 30 hr and with e-folding times also measured in hours. It may be considered, therefore, that in the region immediately over the threshold the instability is dominated by the xenon effect. For values of $\alpha_m T_m^*$ in excess of 13×10^{-3} , however, the motion is purely exponential.

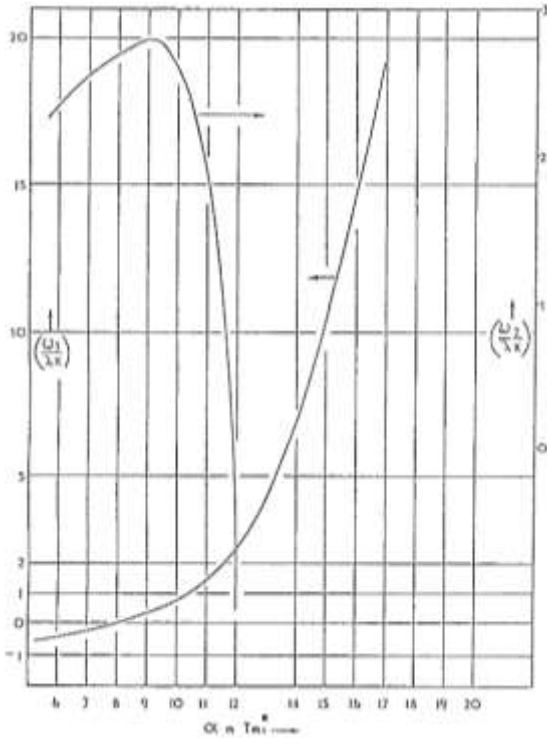


Fig. II-15. Time constants of perturbed motion for the case $M^2 / R_0^2 = 1.7 \times 10^{-3}$, $\alpha_f T_{f1}^* = -5 \times 10^{-3}$

Temperature Effects Alone

For a short while after full power is attained in a reactor there may be little or no ^{135}Xe present. The state of the reactor is then best described by the foregoing equations without the xenon terms. In these conditions the condition for stability becomes

$$M^2 \mu_n - (\alpha_f T_{fn}^* + \alpha_m T_{mn}^*) > 0.$$

The critical value of $\alpha_m T_m^*$ for "temperature only" instability is close to (but slightly greater than) the value of $\alpha_m T_m^*$ at which in the general case the unstable motion changes from oscillatory to exponential form. Thus in the general case, the region of purely real time constant may be referred to as the region of "temperature instability".

For a typical Magnox reactor situation ($\alpha_m = 15 \times 10^{-5} / ^\circ\text{C}$, $T_m^* = 130^\circ\text{C}$), the time constant for fundamental harmonic perturbations is of the order of a few minutes. For the first azimuthal harmonic the equivalent value is about 20 min. In these circumstances the uncontrolled reactor is so far over the stability threshold that even when the xenon is present its effect may be safely neglected.

Control of Spatial Instabilities

The preceding highly approximate analysis has given an indication of how the broad stability characteristics of an uncontrolled reactor may be assessed. With uranium fuel at a reasonably high irradiation, uncontrolled graphite-moderated reactors are certainly unstable to fundamental harmonic disturbances and almost certainly to higher harmonic perturbations also. Predictions of the actual time constants and spatial characteristics of the likely unstable harmonics provide

invaluable information for the control engineer whose job it is to design a control system capable of providing an overall, stable reactor system.

Thus to deal with fundamental harmonic disturbances, which may grow at a rate measured in terms of a few minutes, a reasonably rapidly-acting system is required. To deal with higher harmonic (spatial) disturbances it is of importance to apply the control at the correct position in the reactor. This should clearly be at the points where the harmonics have their greatest amplitudes. The relevant spatial harmonics are illustrated in Fig. II-16. To control these harmonics the reactors may be divided into a number, of sub-regions as shown in Fig. II-17, with each region controlled independently, the control being applied at the center. Providing that it is sufficiently rapidly acting, such a control system will, of course, deal also with fundamental (total power) disturbances. In practice the system operates automatically and control rods are moved on the basis of disturbances in the temperature of the local coolant outlet gas.

Having now designed a suitable control system we may extend the previous harmonic analysis by the inclusion of suitable auxiliary equations which describe the control system. We may then confirm the overall stability of the controlled reactor and indeed investigate the reactor behavior under assumed faulty controller operation - such as the situation when control in one sub-region is inoperative.

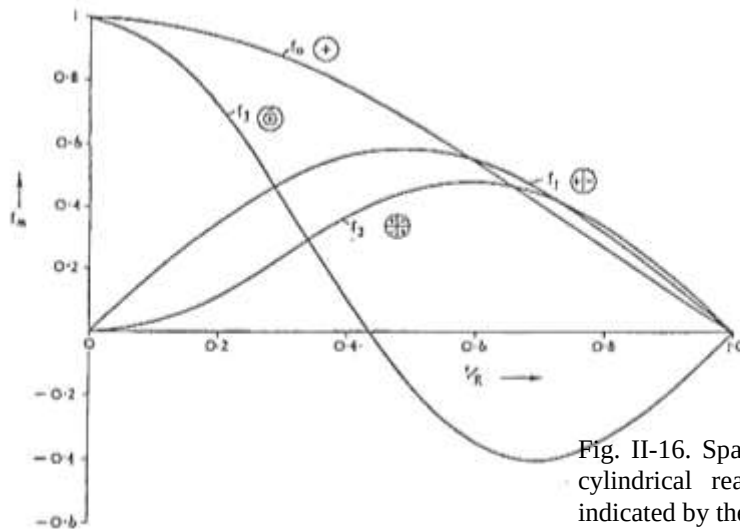


Fig. II-16. Spatial radial harmonics for a uniform cylindrical reactor. The azimuthal variation is indicated by the circular diagram.

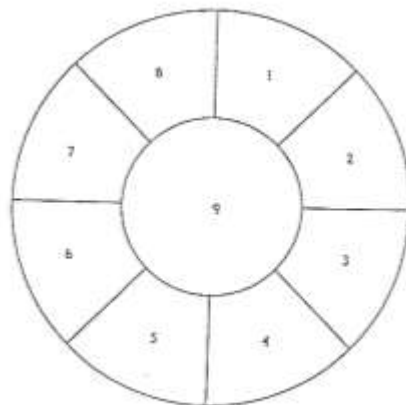


Fig. II-17. Schematic illustration of 9-sector control system